Project Title: *Visualizing Housing Market Trends using Tableau & Flask* **Project Design Phase-I - Solution Fit Template** **Team ID:** PNT2022TMIDxxxxxx

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

>Real estate analysts

>Property managers

>Real estate companies

>Property investors

>Housing market researchers

6. CUSTOMER CONSTRAINTS CC

>Lack of clear visualization tools

>Limited technical knowledge for advanced analytics

>Large and complex housing datasets

>Time-consuming manual analysis

>Difficulty identifying key price-influencing factors

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Explore AS, differentiate

2. JOBS-TO-BE-DONE / PROBLEMS J&P

- *Analyze housing price trends*
- *Understand factors affecting house prices*
- *Compare housing features (bedrooms, bathrooms, floors)*
- *Study renovation impact on property prices*
- *Identify patterns in housing datasets*

9. PROBLEM ROOT CAUSE RC

- *Housing datasets contain many variables*
- *Lack of proper visualization tools*
- *Difficulty analyzing large datasets manually*
- *Limited understanding of market patterns*

7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Identify strong TR & EM

3. TRIGGERS TR

- ***Increasing housing market competition Need for accurate property price prediction Clients asking for housing market insights Large housing datasets available for analysis***
- ***Demand for better data visualization tools.***

4. EMOTIONS: BEFORE / AFTER EM

- *Before*
  - *Confused about housing price pattern*
  - *Frustrated with complex datasets*
  - *Uncertain about property investment decisions*

a) *After*

- *Confident in analyzing housing trends*
- *Better understanding of housing market insights*

10. YOUR SOLUTION SL

- *Develop a Flask-based web application that integrates Tableau dashboards to visualize housing market data.*
- *The system analyzes housing features such as bedrooms, bathrooms, floors, renovation status, and house age and provides interactive visual insights to help real estate professionals make better decisions.*

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

Identify strong TR & EM